

Python Assignment - 1

Theory:

1. What is Python? List any 3 features.

Ans Python is a high-level, interpreted, general-purpose programming language known for its simplicity and readability.

Any 3 features are.

High-Level Language
Large Standard Library
Interpreted.

2. Explain mutable and immutable data types with examples.

Ans mutable data type - This can be changed after creation

ex - my_list = [1, 2, 3]

Immutable data types - This can't be changed after creation

ex - my_tuple = (1, 2, 3)

3. Differentiate between == and is

Ans

==

is

== checks value equality

is checks identities wheather two

wheather two object have same object

variable point to the same memory location

ex -
A = [1, 2]
B = [1, 2]
print (a == b)

output - True

ex -
A = [1, 2]
B = [1, 2]
print (a is b)

output False

4. What is type casting? Explain Implicit and explicit casting

Ans Casting mean converting one data type into another

Implicit Casting - automatically done by python

ex -

explicit Casting - manually done by programmer

5. Difference between single-line and multi-line comments.

ns Single comment - start with # and is used for Short comment

multi-line comment - enclosed is triple quotes

Coding:-

1. Print your name, age and city on separate lines

```
name: "Soubhagya"  
age: 20  
city: "Patna"
```

```
Print (name)  
Print (age)  
Print (city)
```

output

```
Soubhagya  
20  
Patna
```

2. Input two numbers and print sum, difference and product.

```
a = int(input("Enter a number"))  
b = int(input("Enter a number"))  
Print ("Sum: " + str(a+b))  
Print ("difference: ", a-b)  
Print ("Product " + str(a*b))
```

3. Input a number and print whether it is even or odd

```
num = int(input("Enter a number"))  
if num % 2 == 0:  
    Print ("Even")  
else:  
    Print ("Odd")
```



```
name="Soubhagya"  
age=20  
city="Patna"
```

```
print(name)  
print(age)  
print(city)
```



```
Soubhagya  
20  
Patna
```

```
a=int(input("Enter a number"))  
b=int(input("Enter a number"))  
print("sum:a+b")  
print("difference:a-b")  
print("product:a*b")
```

```
Enter a number10  
Enter a number20  
sum:a+b  
difference:a-b  
product:a*b
```

```
num=int(input("Enter a number"))  
if num%2==0:  
    print("Even")  
  
else:  
    print("odd")
```

```
Enter a number20  
Even
```